

Boning Tan 谭泊宁

MASTER'S STUDENT · TONGJI UNIVERSITY
tanews@163.com · tanboning1118.github.io · GitHub

EDUCATION

- 2026.09 — Present

Tongji University
Master's student · Mechanical Engineering
School of Mechanical Engineering and Robotics.
- 2022.09 — 2026.06

Tongji University
B.Eng. · Intelligent Manufacturing Engineering
GPA: 4.70 / 5.00 · CET-6: 543

EXPERIENCE

- 2026.06 — 2026.08

Tencent Robotics X
Data & Evaluation Team Intern
Worked on kinematic singularity analysis for teleoperation measurement arms and ROS-based sensor data acquisition, synchronization, and recording.

PUBLICATION

Guang Chen, Binyao Zhang, **Boning Tan**, Weizhong Yuan. Electric Eel-Inspired Elastomer-Encapsulated Hydrogel Sensor with Superior Anti-Swelling, Self-Adhesion, and Electrical Stability for Diverse Underwater Applications. *Small*, 21(34), 2025. doi:10.1002/sml.202503207
My contribution: CNN-based classification of hydrogel sensor signals for recognition of four swimming postures.

Selected projects

- Underwater soft knee exoskeleton**

2025.12 — 2026.05

- Designed a single-motor bidirectional winch and a compliant following mechanism; used TPU/PLA fabrication to address cable instability.
 - Designed a distribution PCB with 24 V power protection and filtering; integrated four IMUs and CAN communication.
 - Completed waterproof sealing and full-system assembly.
- Six-DOF robot arm simulation**

2024.09 — 2025.01

- Built a classical DH model and derived analytical forward and inverse kinematics.
 - Compared polynomial, LSPB, Bézier, and B-spline trajectories under velocity and acceleration constraints.
 - Implemented feedforward dynamics compensation and PD tracking in Simulink.
- Radar-based human presence sensing**

2024.11 — 2024.12

- Integrated an LD2410 millimeter-wave radar to detect presence, distance, and moving versus stationary targets.
 - Uploaded structured data to OneNet through MQTT, with periodic reporting and automatic reconnection.
- FPGA digital combination lock**

2024.06 — 2024.07

- Designed a six-state controller for password setup, verification, alarms, and timed lockout.
 - Integrated keypad and buzzer feedback, with a 13-bit LFSR for pseudorandom sequences.

SELECTED HONORS

- National First Prize · Chinese College Students Mechanical Engineering Innovation and Creativity Competition, 2025
- Tongji University Undergraduate Outstanding Student Scholarship · 2022–2023, 2023–2024, 2024–2025
- Shanghai Selection Second Prize · Mechanical Engineering Innovation and Creativity Competition, 2024

TECHNICAL TOOLKIT

Python · MATLAB / Simulink · ROS · CNN · CAD / 3D printing · PCB / Embedded systems
Updated: 2026-10-07